

Fine tune LLM and test n8n AI

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Now: qwen3.5 27B(測試模型完畢)

1.生成測試集→2.微調模型→3.測試模型→4.評估分數

Training: user query+1963 templates workflow json

Testing: 1963 user query

	qwen2.5 7B-fine tune (國網中心)	qwen3.5 27B-fine tune (國網中心)
Valid Samples	572/1963	1112/1963
Node Type Accuracy(F1)	0.71	0.363
Connection Accuracy(F1)	0.405	0.112
Parameter Accuracy	0.5	0.317

Fine-tune的資料

初始模型	Fine-tune模型的檔案	instruction	input	output
qwen2.5 7B	n8n_training_data_final.jsonl	請根據需求，生成節點與連線皆無缺失且能成功執行的n8n workflow JSON	{Template原始描述} 長度為61~18475字元 平均為2677字元(英文)	Template workflow json file
qwen3.5 27B	n8n_training_data_v2_humanized	請根據需求，生成節點與連線皆無缺失且能成功執行的n8n workflow JSON	{由qwen3.5 27B閱讀Template後產生的user query} 長度為12~56字元 平均為31字元(中文)	Template workflow json file

Template description v.s LLM 產生的User query(以我資料中的第1個template為例)

Template description:(3584個字元)

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# Auto-create Tiktok Videos with VEED.io AI avatars, ElevenLabs & GPT-4

# 🚀 Viral Tiktok Video Machine: Auto-Create Videos with Your AI Avatar

![[Workflow Overview]](https://www.dr-firas.com/workflow-veed.png)

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### 🧐 Who is this for?
This workflow is for content creators, marketers, and agencies who want to use Veed.io's AI avatar technology to produce viral content. It's ideal for creators who want to appear on camera without recording themselves, and for teams managing multiple brands who need a scalable solution.

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### 🎯 What problem this workflow solves
Manually creating videos for TikTok can take hours – finding trends, writing scripts, recording, and editing. By combining Veed.io, ElevenLabs, and GPT-4, this workflow transforms a simple Telegram input into a ready-to-post video in minutes.

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### 🛠️ What this workflow does
This automation links Veed.io's video-generation API with multiple AI tools:


- Analyzes TikTok trends via Perplexity AI
- Writes a 10-second viral script using GPT-4
- Generates your voiceover via ElevenLabs
- Uses Veed.io (Fabric 1.0 via FAL.ai) to animate your avatar and sync the lips to the voice
- Creates an engaging caption + hashtags for TikTok virality
- Publishes the video automatically via Botatato TikTok API
- Logs all results to Google Sheets for tracking


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### ⚙️ Setup
1. Telegram Bot

- Create your bot via [Botfather](https://t.me/Botfather)
- Configure it as the trigger for sending your photo and theme


2. Connect Veed.io

- Create an account on [Veed.io](https://www.veed.io)
- Get your FAL.ai API key (Veed Fabric 1.0 model)
- Use HTTPS image/audio URLs compatible with Veed Fabric


3. Other APIs

- Add Perplexity, ElevenLabs, and Botatato TikTok keys
- Connect your Google Sheet for logging results

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## 🌟 How to customize this workflow
- Change your Avatar: Upload a new image through Telegram, and Veed.io will generate a new talking version.
- Modify the Script Style: Adjust the GPT prompt for tone (educational, funny, storytelling).
- Adjust Voice Tone: Tweak ElevenLabs stability and similarity settings.
- Expand Platforms: Add Instagram, YouTube Shorts, or X (Twitter) posting nodes.
- Track Performance: Customize your Google Sheet to measure your most successful Veed.io-based videos.

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## 🎯 Expected Outcome
In just a few seconds after sending your photo and theme, this workflow – powered by Veed.io – creates a stream of viral short videos, made without cameras, editing, or effort.

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✅ Import the JSON file in n8n, add your API keys (including Veed.io via FAL.ai), and start generating viral content!

## 📺 [Watch This Tutorial](https://youtu.be/E0e7hiresHk)

![[VEED Logo]](https://www.dr-firas.com/VEED.png)

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📄 Documentation: [Notion Guide](https://automatisation.notion.site/Viral-TikTok-Video-Machine-Auto-Create)

## 🆘 Need help customizing?
Contact me for consulting and support : [LinkedIn](https://www.linkedin.com/in/dr-firas/) / [Youtube](https://www.youtube.com/c/DrFiras)
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LLM產生的User query: (28個字元)幫我用 AI 虛擬人設自動生成並發布爆紅的 TikTok 影片！

為什麼qwen3.5 27b fine tune的結果，分數更低？

1. 仍存在許多格式上的錯誤

2. input大幅縮短(平均2677字元→平均31字元)

3. AI生成的Workflow跟原始Template相比精簡許多

→Template 中包含了大量的 Sticky Note(便條紙)

→Template中會出現一些抓錯用的通知節點 (如telegram)，但AI只會生成跟核心功能有關的節點，導致連線分數(Connection Accuracy)下降

4. 目前的評分程式邏輯為：

Node Type Accuracy--節點類型和數量都會1對1比對，順序則沒有影響

Connection Accuracy--連線的兩端節點類型必須完全正確

Parameter Accuracy--針對相同類型的節點作檢查，比較兩者填的參數，漏填或亂填會降低分數

結論：因為AI只會生成核心功能，不像 template中會加入檢查或一些說明，導致評分程式失準

用以下三個Template當作說明範例

Template 1: Hard(>23個nodes)

Template 2: Medium(15~23個nodes)

Template 21: Easy(≤ 14 個nodes)

V.S

n8n雲端版AI

AI生成與原始template的差異(template 1)

原始的template:

VIRAL TIKTOK CREATOR - SETUP GUIDE

Watch This Tutorial



This workflow transforms a photo + theme into viral TikTok content automatically.

WHAT YOU NEED:

- Telegram Bot Token
- ElevenLabs API Key + Voice ID
- veed.io API Key with FAL.ai
- Google Sheets Document
- OpenAI API Key
- Perplexity API Key

Setup

- Import this workflow into your n8n instance.
- Connect your Google Sheets, Gmail, and Google Calendar credentials. Tutorial: [Configure Your Google Sheets, Gmail, Calendar Credentials](#)
- Connect Your Data in Google Sheets. Data must follow this format: [Sample Sheets Data](#)

Need Help or Want to Customize This?

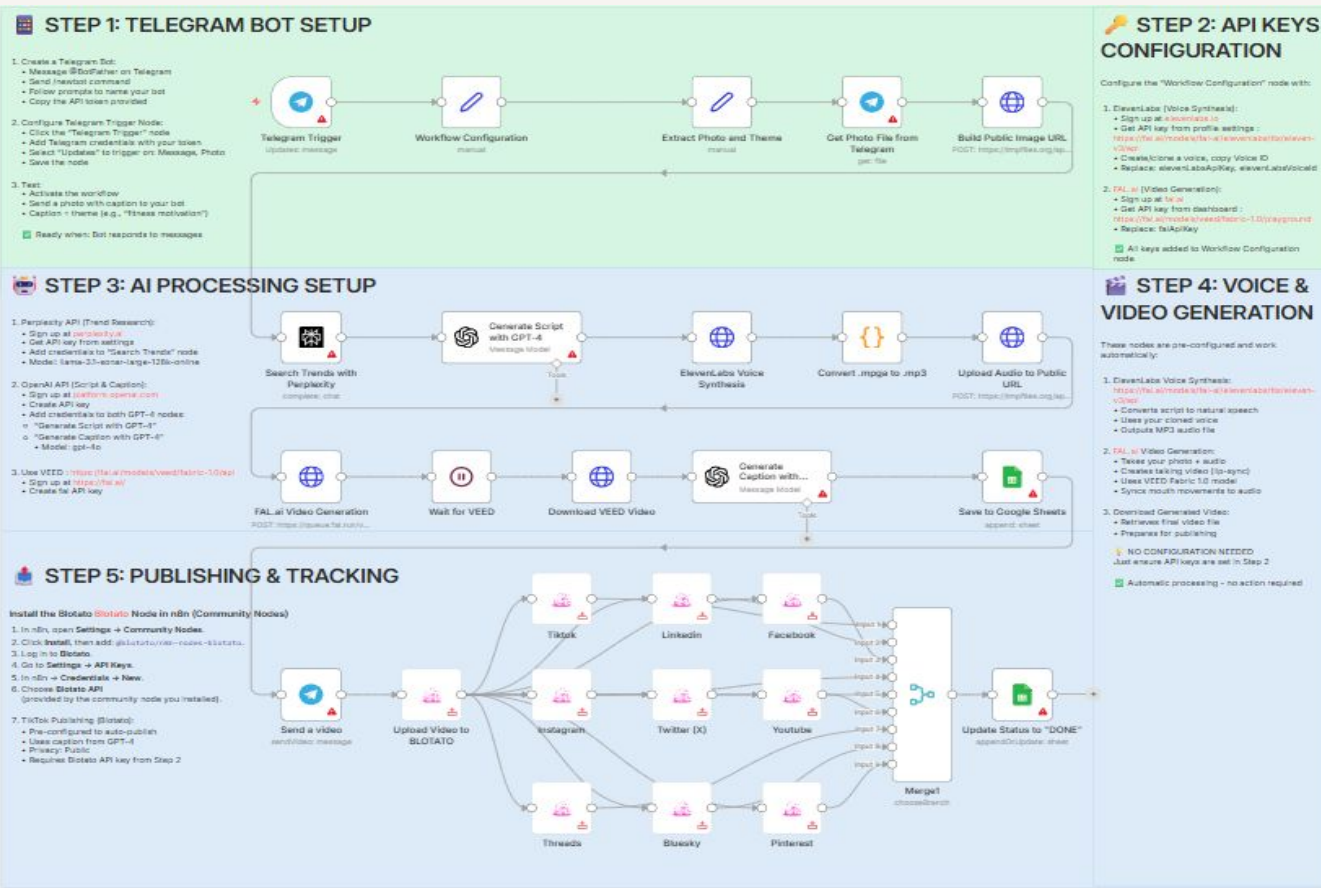
Contact me for consulting and support: [LinkedIn](#) / [YouTube](#)

HOW IT WORKS - WORKFLOW FLOW

- You send: Photo + Theme via Telegram
- Perplexity finds current viral trends
- GPT-4 creates: 10-second viral script
- ElevenLabs generates: Voice audio (MP3)
- FAL.ai creates: Talking video from photo
- GPT-4 writes: Optimized caption + hashtags
- Saves to: Google Sheets (tracking)
- Publishes to: TikTok automatically

TOTAL TIME: ~2-3 minutes
RESULT: Viral TikTok video published!

USAGE:
Send photo with caption "Fitness tips"
→ Get viral fitness video on TikTok



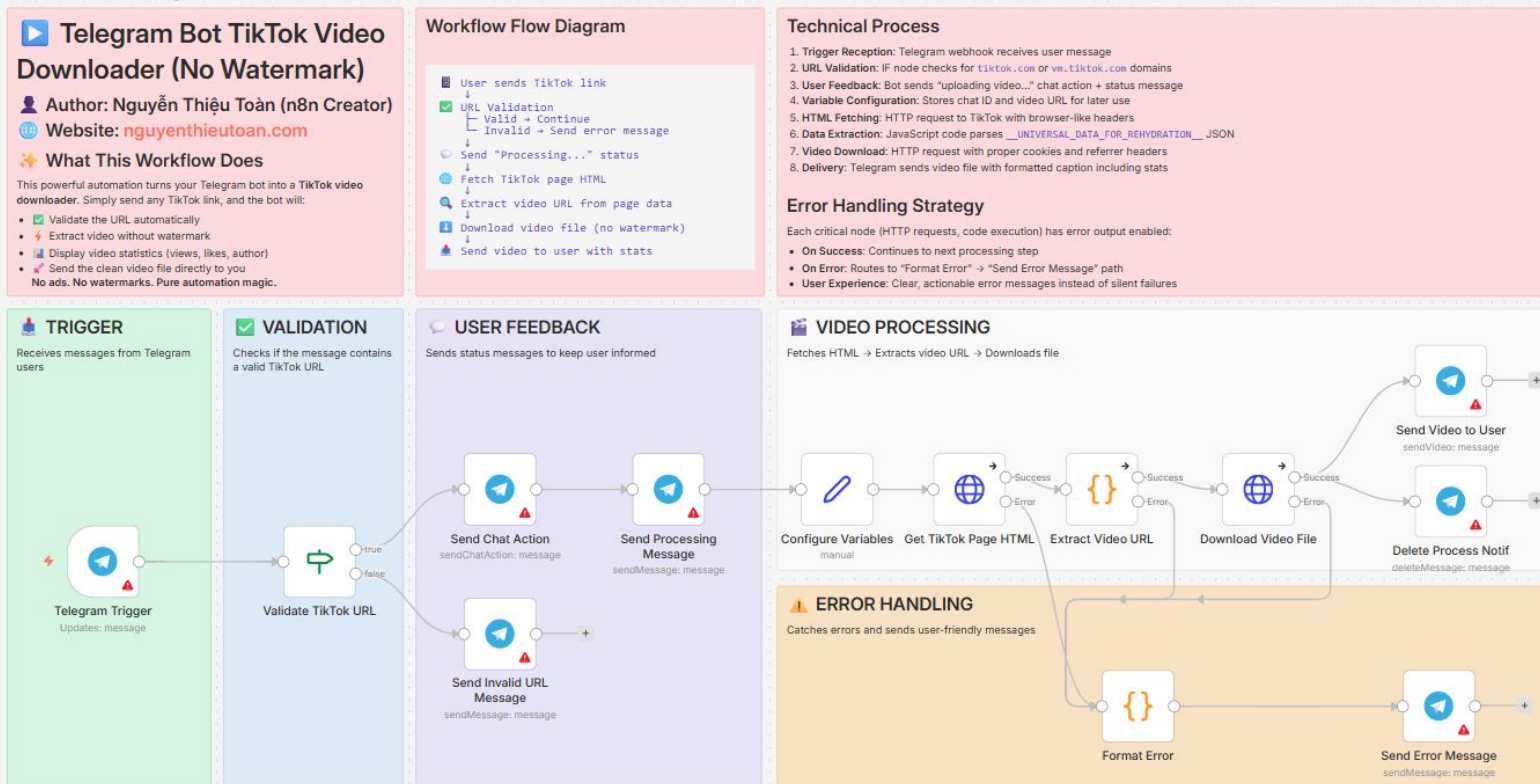
AI生成與原始template的差異(template 1)

n8n雲端版的AI: (input: 幫我用 AI 虛擬人設自動生成並發布爆紅的 TikTok 影片！)



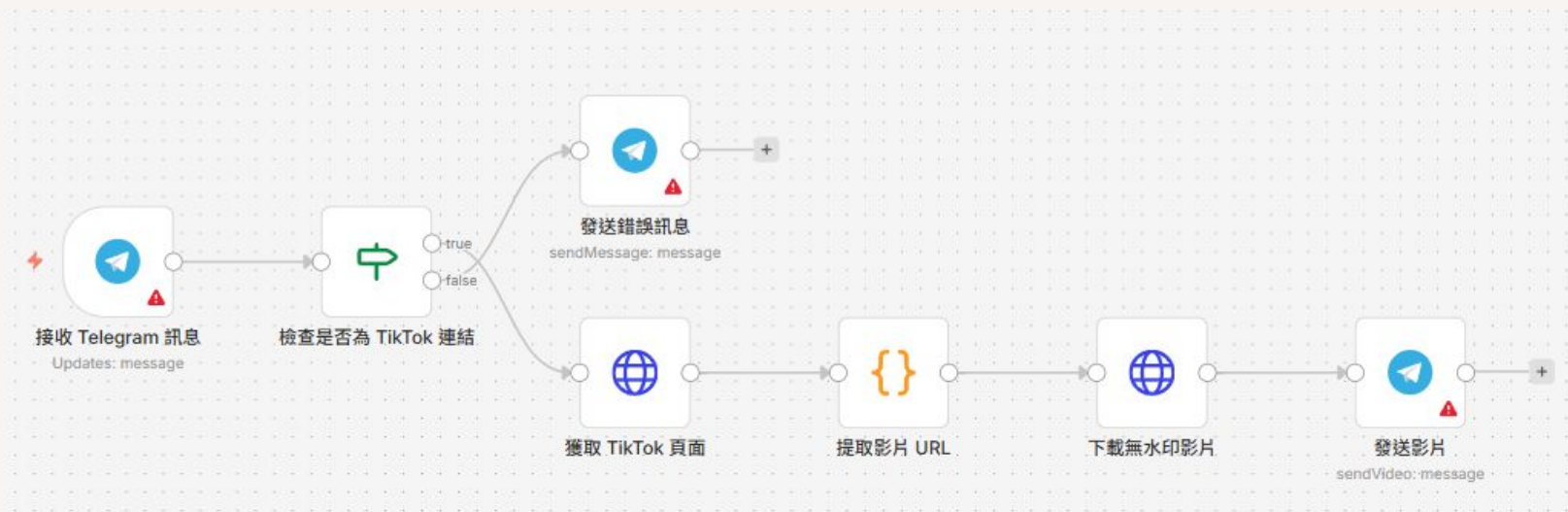
AI生成與原始template的差異(template 2)

原始的template:



AI生成與原始template的差異(template 2)

n8n雲端版的AI: (input: 幫我做個 Telegram 機器人, 傳連結就能下無水印的 TikTok 影片。)



AI生成與原始template的差異(template 21)

原始的template:

For more tutorials visit our [Youtube](#)

1. Setup

- Create a Telegram bot (via @BotFather)
 - Copy your Bot Token into the Telegram nodes.
- Add credentials in n8n:
 - Telegram (Telegram Trigger, Download File)
 - OpenAI (API key, for Transcribe Audio, LinkedIn Post Text, Image Prompt and Create Image)
 - LinkedIn (OAuth - Create a post)
 - For self-hosted n8n: Create LinkedIn app at [developer.linkedin.com](#) to obtain OAuth credentials (Client ID and Client Secret)
- Activate the workflow
 - Send your text or voice message via Telegram bot.
 - OpenAI modules will create optimized LinkedIn Post including an image and post it on LinkedIn.

2. How to customize this workflow to your needs

- Modify node logic:
 - Change LinkedIn Post Text prompt to match your writing style
 - Adjust image generation settings (size, style, model) in Create Image node
- Add integrations:
 - Connect Google Sheets, Airtable, or Notion for post approval workflow
 - Add scheduling nodes for delayed publishing
 - Track analytics by saving post data to database
- Multi-platform posting:
 - Repost on Twitter, Facebook, Instagram

3. Node Breakdown & Flow

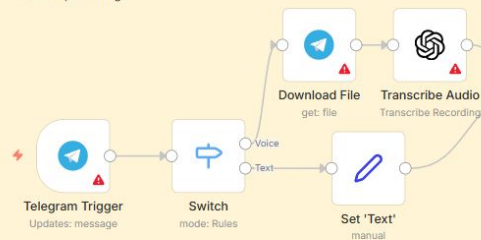
- Telegram Trigger: Listens for all messages.
- Switch: Routes messages (voice, text) to the right logic.
- Voice: Download File → Transcribe Audio → LinkedIn Post Text
- Text: → LinkedIn Post Text directly
- LinkedIn Post Text: Formats content into professional post with proper structure
- Image Prompt: Generates detailed image description based on post content
- Create Image: Produces AI visual matching post topic
- Create a post: Posts text and image on LinkedIn platform

Need Help or Want to Customize?

For further automation or integration, feel free to [book an appointment](#) or [LinkedIn](#)

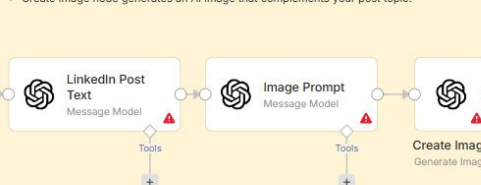
1. Message Input

- Telegram Trigger receives voice or text messages from your bot. Switch node routes each type to the correct path.
- Download File retrieves audio from Telegram. Transcribe Audio (OpenAI) converts speech to text for processing.



2. AI Content Generation

- LinkedIn Post Text (OpenAI) takes your raw input and transforms it into a professional LinkedIn post.
- Image Prompt node analyzes your post content and creates a detailed visual description.
- Create Image node generates an AI image that complements your post topic.



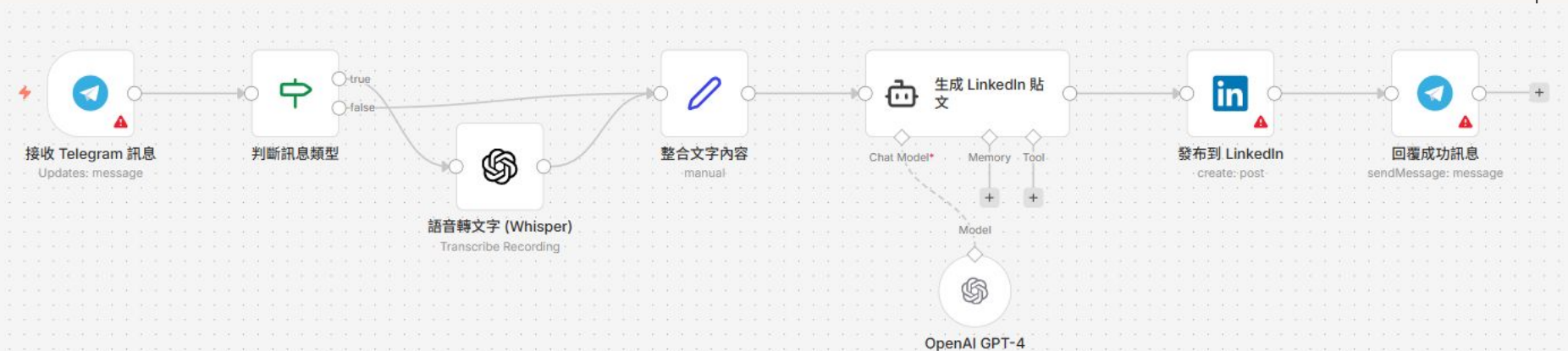
3. Post on LinkedIn

- Double-click "Create a post" node for OAuth authentication
- Self-hosted users: Create LinkedIn app at [developer.linkedin.com](#) to get Client ID/Secret
- Once connected, posts automatically publish with text and AI image



AI生成與原始template的差異(template 21)

n8n雲端版的AI: (input: 用 Telegram 傳文字或語音, 就能自動生成並發布專業 LinkedIn 貼文 !)

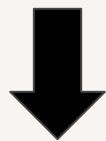


評分結果

Index	需求摘要	原始template 難度級別	Node Type (F1)	Connection (F1)
1	用 AI 虛擬人設生成發布 TikTok 影片	Hard (28節點)	0.316	0.154
2	傳連結下載無水印 TikTok 影片的機器人	Medium (21節點)	0.381	0.118
21	傳語音自動生成發布 LinkedIn 貼文	Easy (9節點)	0.588	0.000

加長prompt來測試n8n AI

幫我每天自動分析生意數據，用 AI 算出關鍵指標並推播建議給我。



請建立一個 n8n workflow，每天定時自動分析多來源業務數據。Schedule Trigger 觸發後，同時從 Microsoft SQL Server 查詢銷售與庫存數據(兩個 SQL 節點)，並從 Google Analytics 抓取網站流量；用多個 IF 節點判斷各指標是否異常(如銷售下滑 > 10%、庫存不足)；summarize 節點彙整各類數據、set 節點整理輸出格式；Google Sheets 記錄每日趨勢；最後用 emailSend 節點發送含 AI 關鍵指標分析與建議的每日報告給管理層。

	Short Prompt	Long Prompt
平均長度	29.7 字元 / 3.3 詞	261.3 字元 / 23.5 詞
比例	1x	8.8x

評分結果

Template 619(Easy), 1567(Easy), 246(Medium), 1125(Medium), 142(Hard), 287(Hard)

619 short prompt doesn't come out result, the grades doesn't include it

	qwen3.5 27b(short prompt)		n8n AI (short prompt)		n8n AI (long prompt)	
Difficulty	Node F1	Conn F1	Node F1	Conn F1	Node F1	Conn F1
Easy	0.310	0.000	0.615	1.000	0.923	0.611
Medium	0.224	0.040	0.289	0.045	0.616	0.220
Hard	0.264	0.000	0.240	0.000	0.686	0.315
Overall	0.266	0.013	0.335	0.218	0.742	0.382

Thank you